Statistical Mechanics: Exercises

Duy M. Dao

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Scope of the exercises. The exercises are selected from three problem collections: A. A. Schekochihin and J. Devriendt, *Oxford Problem Sets 5–8: Statistical Mechanics* (2017), cited as [S]; D. Cohen, *Exercises in Statistical Mechanics* (Ben-Gurion University, 2019), cited as [C]; and D. A. R. Dalvit, J. Frastai and I. D. Lawrie, *Problems on Statistical Mechanics* (IOP, 1999), cited as [D]. Only problems that the note covers and that require more than a substitution into one formula are included. The problems are rewritten in the notation of the note, and some are merged or extended. Problems on quantum gases, kinetic theory, interacting lattice models and continuous transitions are left out, because the note does not cover them yet. The collections contain no problems on binary phase diagrams, so Section 11 of the note has no exercises here. Each exercise names the related sections of the note and the source, and ends with an answer for checking. Section 7 adds the five problems of Quiz 1 of the phase-diagram course, which cover Section 11 of the note and one ternary system. Full solutions with figures are in the separate file `quiz1.pdf`. Guided exercises on point defects, built from the equations of a review article, are in the separate sheet `Freysoldt/freysoldt.pdf`.

Units and notation. The Boltzmann constant k_B is kept explicit, and $\beta = 1/(k_B T)$. Where a problem needs the classical ideal gas, the canonical partition function of N identical particles of mass m in the volume V is given as $Z_N = V^N/(N!\,\lambda_{\mathrm{th}}^{3N})$, with the thermal wavelength $\lambda_{\mathrm{th}} = h/\sqrt{2\pi m k_B T}$.

1 Entropy Maximization

Exercise 1: Loaded die. A die has the outcomes $i = 1, \ldots, 6$ with unknown probabilities p_i.

(a) With no information beyond normalization, find the MaxEnt distribution and the expected outcome $\langle i \rangle$.

(b) Many throws give the average $\bar{\imath} = 3.667$. Use MaxEnt with the constraint $\langle i \rangle = \bar{\imath}$ to show that $p_i = x^i / \sum_{j=1}^{6} x^j$, with $x = e^{-\lambda}$ for the multiplier λ, and that x solves

$$\sum_{i=1}^{6} (i - \bar{\imath})\, x^i = 0.$$

(c) Solve for x numerically, list the p_i, and compare the entropy with $\ln 6$.

Answer. (a) $p_i = 1/6$ and $\langle i \rangle = 3.5$. (c) $x = 1.059$ ($\lambda = -0.057$), and $p_1, \ldots, p_6 = 0.144$, 0.152, 0.161, 0.171, 0.181, 0.192. The entropy is 1.787, only 0.005 below $\ln 6 = 1.792$, because the constraint is weak.
Related: Sections 1.2 and 3 of the note. *Source:* [S] 5.3.

Exercise 2: Additivity forces the logarithm. Suppose that the entropy of an isolated system is some function $S = \phi(\Omega)$ of the number of accessible microstates, with ϕ increasing.

(a) Two independent systems have Ω_1 and Ω_2 microstates. Show that additivity of the entropy requires $\phi(\Omega_1 \Omega_2) = \phi(\Omega_1) + \phi(\Omega_2)$, and hence $\phi(1) = 0$ and $\phi(\Omega^n) = n\,\phi(\Omega)$ for every positive integer n.

(b) For integers $\Omega, M \geq 2$ and a large integer n, choose the integer m with $M^m \leq \Omega^n < M^{m+1}$. Use the monotonicity of ϕ to show that

$$\left| \frac{\phi(\Omega)}{\phi(M)} - \frac{\ln \Omega}{\ln M} \right| \leq \frac{1}{n},$$

and conclude that $\phi(\Omega) = k \ln \Omega$ with a constant $k > 0$.

(c) Show that the Gibbs–Shannon entropy of a uniform distribution over Ω microstates has this form, and explain why the note sets $k = k_B$.

Answer. (b) Both $\phi(\Omega)/\phi(M)$ and $\ln \Omega / \ln M$ lie in $[m/n, (m+1)/n)$. The ratio $k = \phi(M)/\ln M$ is therefore the same for every M. (c) $S = \ln \Omega$ for $p_i = 1/\Omega$, and k_B only fixes the units.
Related: Sections 1.1 and 2 of the note. *Source:* [C] 1043.

2 The Microcanonical Ensemble

Exercise 3: Two-level system and negative temperature. N distinguishable, non-interacting particles each have the energy $-\varepsilon$ or $+\varepsilon$. The system is isolated with the energy E, so that $n = (N + E/\varepsilon)/2$ particles occupy the upper level.

(a) Find $S(E)$ for $N, n \gg 1$, and sketch S against E.

(b) From $1/T = \partial S/\partial E$, find $T(E)$. Show that $T < 0$ for $E > 0$, and explain why such states are hotter than any state with $T > 0$.

(c) The system at $T < 0$ touches a heat bath at $T_0 > 0$. Use the entropy change $\delta S = k_B(\beta_1 - \beta_2)\,\delta U_1$ of Section 3.4 of the note to find the direction of the energy flow.

(d) Find the entropy change from $T = 0^+$ to $T \to \pm\infty$ and express it in bits.

(e) Explain why an ideal gas in a box cannot reach $T < 0$.

Answer. (a) $S = -Nk_B[x \ln x + (1-x)\ln(1-x)]$ with $x = n/N$, maximal at $E = 0$. (b) $\beta = (2\varepsilon)^{-1}\ln[(N-n)/n] = -(2\varepsilon)^{-1}\ln[(N\varepsilon+E)/(N\varepsilon-E)]$. As E increases, β decreases continuously from $+\infty$ through 0 to $-\infty$. (c) Energy flows from the system to the bath, because the bath has the larger β. (d) $\Delta S = Nk_B \ln 2$, that is, N bits. (e) The kinetic energy has no upper bound, so $S(E)$ increases for all E.
Related: Sections 2 and 3.4 of the note. *Source:* [C] 1047; [D] 3.6.

Exercise 4: Frenkel defects. A crystal has N atoms on N lattice sites and M interstitial sites, with M of order N. Moving an atom from a lattice site to an interstitial site costs the energy ω. Let $x = e^{-\beta\omega}$.

(a) Show that the canonical partition function is

$$Z(x) = \sum_n \binom{N}{n}\binom{M}{n} x^n.$$

(b) Treat n as continuous, with $d\ln n!/dn \approx \ln n$, and find the largest term. Show that $n^2/[(N-n)(M-n)] = x$, and hence $\bar{n} \approx \sqrt{NM}\, e^{-\beta\omega/2}$ for $x \ll 1$.

(c) Interpret the condition of part (b) as a law of mass action for the reaction "lattice atom $\rightleftharpoons$ vacancy + interstitial".

(d) Keep only the largest term in $\ln Z$ and find F, U, S and C_V for $x \ll 1$.

(e) Explain why a Gaussian sum around the largest term changes $\ln Z$ only by $O(\ln \bar{n})$.

Answer. (b) For $N = M = 4000$ and $x = e^{-12}$, the exact mean is 9.64 and the approximation gives 9.92. (d) $\ln Z \approx 2\bar{n}$, so $F = -2k_B T\bar{n}$, $U = \bar{n}\omega$, $S = k_B\bar{n}(2 + \beta\omega)$ and $C_V = 2k_B\bar{n}(\beta\omega/2)^2$. The factor $1/2$ in the exponent of $\bar{n}$ reflects that each defect creates two independent objects, a vacancy and an interstitial.
Related: Sections 2, 3 and 3.3 of the note. *Source:* [C] 2311.

3 The Canonical Ensemble

Exercise 5: Elastic chain and superfluous constraints. Here, τ denotes a tension and Λ the length of a chain. A one-dimensional chain has N segments. Each segment is horizontal, with the length a and the energy 0, or vertical, with the length 0 and the energy τa, where τ is a fixed tension.

(a) Find the canonical partition function and the mean energy $U(T)$.

(b) Show that the mean length is $\langle\Lambda\rangle = Na - U/\tau$. For $\beta\tau a \ll 1$, show that $\tau = Ak_B T(\langle\Lambda\rangle - \Lambda_0)$, a Hooke law, and find A and Λ_0. Explain why a hotter chain needs a larger tension for the same length.

(c) Find the heat capacity at fixed τ, and explain why it vanishes at low and at high temperature.

(d) Find the condition on U for $T < 0$.

(e) Maximize the entropy with two constraints, $\langle E\rangle = U$ and $\langle\Lambda\rangle = \Lambda$, with the multipliers β and σ. Show that a solution exists only if $\Lambda = Na - U/\tau$, and that only the combination $\beta - \sigma/\tau$ is determined. Relate this result to the uniqueness condition for the multipliers in Section 5 of the note.

Answer. (a) $Z = (1 + e^{-\beta\tau a})^N$ and $U = N\tau a/(e^{\beta\tau a} + 1)$. (b) $\Lambda_0 = Na/2$ and $A = 4/(Na^2)$. The tension opposes the entropy, which favors the disordered half-and-half state, and the entropic restoring force grows with T. (c) $C = Nk_B(\beta\tau a)^2 e^{\beta\tau a}/(e^{\beta\tau a} + 1)^2$, the Schottky form, with a maximum near $k_B T \approx 0.42\,\tau a$. (d) $U > N\tau a/2$. (e) Every microstate satisfies $E + \tau\Lambda = N\tau a$, so the covariance matrix of (E, Λ) is singular.
Related: Sections 1.2, 3 and 5 of the note. *Source:* [S] 5.5.

Exercise 6: Zipper model and the thermodynamic limit. A double-stranded molecule has N links that open only in sequence from one end, and the last link never opens. An open link costs the energy ε and has g orientations. A state with p open links has the energy $p\varepsilon$ and the degeneracy g^p.

(a) With $x = g\,e^{-\beta\varepsilon}$, show that $Z = (1 - x^N)/(1 - x)$.

(b) Find $\langle p\rangle$, and show that $\langle p\rangle/N \to 0$ for $x < 1$ and $\langle p\rangle/N \to 1$ for $x > 1$ as $N \to \infty$.

(c) Show that $\langle p\rangle \approx (N-1)/2 + (x-1)(N^2-1)/12$ near $x = 1$, and find the transition temperature T_c.

(d) Find the heat capacity at $x = 1$ and show that it grows as N^2.

(e) Explain why the free energy is analytic in T for every finite N, and why a transition nevertheless appears in the limit. Which order is the transition?

Answer. (b) $\langle p\rangle = x/(1-x) - Nx^N/(1-x^N)$. For $N = 500$, $\langle p\rangle/N = 0.04$, 0.50 and 0.96 at $x = 0.95$, 1 and 1.05. (c) $T_c = \varepsilon/(k_B \ln g)$. The slope is the variance $(N^2-1)/12$ of a uniform distribution. (d) $C = k_B(\ln g)^2(N^2-1)/12$. (e) Z is a polynomial in x, so $\ln Z$ is analytic for finite N. In the limit, the fraction of open links jumps from 0 to 1 with the latent heat ε per link, a first-order transition.
Related: Sections 3 and 10 of the note. *Source:* [C] 2360.

Exercise 7: Free energy of a perturbed Hamiltonian. A classical system has the Hamiltonian $\mathcal{H} = \mathcal{H}_0 + \lambda\mathcal{H}_1$. Let F_0 and $\langle\cdot\rangle_0$ denote the free energy and the averages at $\lambda = 0$.

(a) Show that $e^{-\beta(F-F_0)} = \langle e^{-\beta\lambda\mathcal{H}_1}\rangle_0$.

(b) Expand in λ and show that

$$F = F_0 + \lambda\langle\mathcal{H}_1\rangle_0 - \frac{\beta\lambda^2}{2}\,\mathrm{Var}_0(\mathcal{H}_1) + \frac{\beta^2\lambda^3}{6}\,\kappa_3 + O(\lambda^4),$$

where $\kappa_3 = \langle(\mathcal{H}_1 - \langle\mathcal{H}_1\rangle_0)^3\rangle_0$.

(c) Show that $\partial^2 F/\partial\lambda^2 = -\beta\,\mathrm{Var}_\lambda(\mathcal{H}_1) \le 0$ for every λ, and deduce the Gibbs–Bogoliubov bound $F \le F_0 + \lambda\langle\mathcal{H}_1\rangle_0$.

(d) Find $U = \langle\mathcal{H}\rangle$ to first order in λ.

Answer. (b) The expansion is the cumulant expansion of $\ln\langle e^{-\beta\lambda\mathcal{H}_1}\rangle_0$. (c) $F(\lambda)$ is concave, so the curve lies below the tangent at $\lambda = 0$. (d) $U = U_0 + \lambda\,[\langle\mathcal{H}_1\rangle_0 - \beta\,\mathrm{Cov}_0(\mathcal{H}_0, \mathcal{H}_1)] + O(\lambda^2)$.
Related: Sections 3 and 3.3 of the note. *Source:* [D] 3.13.

4 Thermodynamic Potentials and Legendre Transformations

Exercise 8: Heat capacities and compressibilities. Here, $\alpha_P = V^{-1}(\partial V/\partial T)_P$ is the thermal expansivity, $\kappa_T = -V^{-1}(\partial V/\partial P)_T$ the isothermal compressibility and $\kappa_S = -V^{-1}(\partial V/\partial P)_S$ the adiabatic compressibility, all at fixed N.

(a) From the equality of the mixed second derivatives of $F(T, V, N)$ and of $G(T, P, N)$, derive the Maxwell relations $(\partial S/\partial V)_T = (\partial P/\partial T)_V$ and $(\partial S/\partial P)_T = -(\partial V/\partial T)_P$.

(b) Show that $T\,dS = C_V\,dT + T(\alpha_P/\kappa_T)\,dV$ and $T\,dS = C_P\,dT - TV\alpha_P\,dP$.

(c) Prove that

$$C_P - C_V = \frac{TV\alpha_P^2}{\kappa_T}, \qquad \frac{C_P}{C_V} = \frac{\kappa_T}{\kappa_S}.$$

(d) Express C_P and C_V in terms of T, V, α_P, κ_T and κ_S.

(e) Use the stability conditions of the note to show that $C_P \geq C_V > 0$ and $\kappa_T \geq \kappa_S > 0$.

Answer. (b) $(\partial P/\partial T)_V = \alpha_P/\kappa_T$ by the cyclic rule. (d) $C_P = TV\alpha_P^2/(\kappa_T - \kappa_S)$ and $C_V = TV\alpha_P^2\kappa_S/[\kappa_T(\kappa_T - \kappa_S)]$. (e) $C_V > 0$ and $\kappa_T > 0$ follow from the concavity of S and the convexity of F in V. Then $C_P - C_V \geq 0$, and $\kappa_S = \kappa_T C_V/C_P$.
Related: Sections 3.3, 4.4 and 5.3 of the note. *Source:* [D] 2.4.

Exercise 9: Rubber band in the constant-tension ensemble. A polymer has N monomers on a line. Each monomer has the length a and the energy E_a, or the length b and the energy E_b, with $a > b$. The polymer is held at the temperature T under the tension τ.

(a) Maximize the entropy with the constraints $\langle E\rangle$ and $\langle\Lambda\rangle$, identify the multiplier of the length as $-\beta\tau$, and show that

$$\Gamma(T, \tau) = \left[e^{-\beta(E_a - \tau a)} + e^{-\beta(E_b - \tau b)}\right]^N.$$

(b) Show that $G(N, \tau, T) = -k_B T \ln\Gamma = F - \tau\Lambda$ is the Legendre transform of $F(N, \Lambda, T)$ in Λ. Name the ensemble by the natural variables of G, and find $\Lambda = -\partial G/\partial\tau$.

(c) Find $\Lambda(T \to \infty, \tau)$. For $E_a = E_b$ and weak tension, show that $\tau = 4k_BT(\Lambda - \Lambda_\infty)/[N(a-b)^2]$.

(d) At $\tau = 0$ and $E_a = E_b$, find the distribution $P(\Lambda)$ of the length for large N. Show that the entropic force $\tau = -k_BT\,\partial \ln P/\partial\Lambda$ reproduces part (c).

(e) Let each monomer take any length in $[b, a]$ with equal energy. Find Γ and $\Lambda(T, \tau)$.

Answer. (b) The $N\tau T$ ensemble, with $\Lambda = N\,[b + (a-b)\,p_a]$ and $p_a = \{1 + e^{\beta[(E_a - \tau a) - (E_b - \tau b)]}\}^{-1}$. (c) $\Lambda_\infty = N(a+b)/2$. (d) $P(\Lambda)$ is Gaussian with the mean Λ_∞ and the variance $N(a-b)^2/4$. (e) $\Gamma = \left[(e^{\beta\tau a} - e^{\beta\tau b})/(\beta\tau(a-b))\right]^N$ and $\Lambda = N\left[(ae^{\beta\tau a} - be^{\beta\tau b})/(e^{\beta\tau a} - e^{\beta\tau b}) - k_BT/\tau\right]$.
Related: Sections 1.2, 4.4 and 4.5 of the note. *Source:* [C] 2351.

5 The Grand Canonical Ensemble

Exercise 10: The grand canonical distribution from a reservoir. A small system exchanges energy and particles with a large reservoir, and the two together are isolated, with the total energy E_{tot} and the total particle number N_{tot}.

(a) Assume equal probabilities for the microstates of the isolated whole. Show that the probability of a system microstate i is proportional to $\Omega_R(E_{\text{tot}} - E_i, N_{\text{tot}} - N_i)$.

(b) Expand $S_R = k_B \ln \Omega_R$ to first order and recover $p_i \propto e^{-\beta(E_i - \mu N_i)}$. Identify T and μ with derivatives of S_R.

(c) Estimate the second-order terms and show that they vanish as the reservoir becomes infinitely large.

(d) Compare this derivation with the MaxEnt derivation in Section 5 of the note. Which assumption replaces the constraints on $\langle E \rangle$ and $\langle N \rangle$?

Answer. (b) $1/T = \partial S_R/\partial E$ and $-\mu/T = \partial S_R/\partial N$. (c) The second derivatives scale as $1/N_R$, so their terms are of order E_i^2/N_R. (d) The reservoir fixes T and μ through the reservoir entropy, whereas MaxEnt fixes the multipliers through the measured averages. Both give the same distribution.
Related: Sections 3.4 and 5 of the note. *Source:* [S] 6.11.

Exercise 11: Energy fluctuations with particle exchange. Here, $z = e^{\beta\mu}$ denotes the fugacity.

(a) From $\ln \Xi(\beta, z)$ at fixed V, show that $\mathrm{Var}_G(E) = -(\partial U/\partial\beta)_z$, $\mathrm{Var}(N) = z(\partial N/\partial z)_\beta$, and $\mathrm{Cov}(E, N) = z(\partial U/\partial z)_\beta = -(\partial N/\partial\beta)_z$.

(b) In the thermodynamic limit, U in the grand canonical ensemble equals $U(\beta, V, N)$ in the canonical ensemble with $N = N(\beta, z)$. Use the chain rule to prove

$$\mathrm{Var}_G(E) = \mathrm{Var}_C(E) + \left(\frac{\partial U}{\partial N}\right)^2_{T,V} \mathrm{Var}(N).$$

(c) Explain why the grand canonical energy fluctuation exceeds the canonical one.

(d) For the monatomic ideal gas, with $U = \frac{3}{2}Nk_BT$ and $\mathrm{Var}(N) = N$, find the ratio $\mathrm{Var}_G(E)/\mathrm{Var}_C(E)$.

Answer. (b) $(\partial U/\partial\beta)_z = (\partial U/\partial\beta)_N + (\partial U/\partial N)_\beta(\partial N/\partial\beta)_z$, and $-(\partial N/\partial\beta)_z = (\partial U/\partial N)_\beta \,\mathrm{Var}(N)$. (c) The exchanged particles carry energy. (d) $\mathrm{Var}_G(E) = \frac{15}{4}N(k_BT)^2$ and $\mathrm{Var}_C(E) = \frac{3}{2}N(k_BT)^2$, so the ratio is $5/2$.
Related: Sections 3.3, 5 and 5.1 of the note. *Source:* [D] 3.28.

Exercise 12: Particle-number fluctuations and compressibility.

(a) Use the Gibbs–Duhem relation at fixed T, $dP = (N/V)\,d\mu$, to show that $(\partial N/\partial\mu)_{T,V} = N^2\kappa_T/V$. Hence express $\mathrm{Var}(N)/N$ through κ_T.

(b) Show that the third cumulant is $\langle (N - \langle N\rangle)^3\rangle = (k_BT)^2(\partial^2 N/\partial\mu^2)_{T,V}$.

(c) For the classical ideal gas, show that $\Xi = \exp(zV/\lambda_{\mathrm{th}}^3)$, that all cumulants of N equal $\langle N\rangle$, and that N has the Poisson distribution $P(N) = \langle N\rangle^N e^{-\langle N\rangle}/N!$.

(d) Explain why $\mathrm{Var}(N)/N$ grows without bound as a fluid approaches the critical point.

Answer. (a) $\mathrm{Var}(N)/N = k_BT\,(N/V)\,\kappa_T$. (b) The nth cumulant is $(k_BT)^{n-1}\partial^{n-1}N/\partial\mu^{n-1}$. (c) $N = zV/\lambda_{\mathrm{th}}^3 \propto e^{\beta\mu}$, so every derivative $k_BT\,\partial/\partial\mu$ returns N. (d) $\kappa_T \to \infty$ at the critical point, where $\partial P/\partial v = 0$.
Related: Sections 5.1 and 5.7 of the note. *Source:* [D] 2.5; [C] 1027; [S] 6.9.

Exercise 13: Langmuir adsorption. A classical ideal gas at the pressure P and the temperature T is in contact with a wall that has N_0 independent sites. Each site is empty, with the energy 0, or holds one molecule, with the energy $-\varepsilon$.

(a) Use the result of Exercise 12 to show that the fugacity of the gas is $z = P\lambda_{\text{th}}^3/(k_BT)$.

(b) Treat the sites as a grand canonical system with the same T and μ as the gas. Find the grand partition function of one site and the mean number of adsorbed molecules $\langle N_s \rangle$.

(c) Write the coverage $\theta = \langle N_s \rangle / N_0$ as $P/(P + P_0(T))$ and find $P_0(T)$. Discuss the limits of low and high pressure.

(d) Find $\text{Var}(N_s)$, and explain why it vanishes at full coverage.

Answer. (b) $\xi = 1 + ze^{\beta\varepsilon}$ and $\langle N_s \rangle = N_0 ze^{\beta\varepsilon}/(1 + ze^{\beta\varepsilon})$. (c) $P_0 = (k_BT/\lambda_{\text{th}}^3)\, e^{-\beta\varepsilon}$. At low pressure $\theta \approx P/P_0$ (Henry's law), and at high pressure $\theta \to 1$. (d) $\text{Var}(N_s) = N_0\theta(1-\theta)$.
Related: Sections 5 and 5.5 of the note. *Source:* [D] 3.25.

Exercise 14: The NPT ensemble of an ideal gas. The partition function of the NPT ensemble in the note is $\Gamma = \sum_i e^{-\beta(E_i + PV_i)}$. For a gas in a container of variable volume, replace the sum over volumes by $\int_0^\infty dV/\Delta V$, where ΔV is a "quantum of volume".

(a) With $Z_N(V)$ of the classical ideal gas, show that

$$\Gamma = \frac{1}{\Delta V\, \lambda_{\text{th}}^{3N}} \left(\frac{k_BT}{P} \right)^{N+1}.$$

(b) Find $G = -k_BT \ln \Gamma$. Show that $G = N\mu$ with $\mu = k_BT \ln(P\lambda_{\text{th}}^3/k_BT)$, up to a term that does not grow with N, and find the condition on ΔV for that term to be negligible.

(c) Find $\langle V \rangle = (\partial G/\partial P)_{T,N}$ and the equation of state.

(d) Find $\text{Var}(V)/\langle V \rangle^2$, and relate it to κ_T.

Answer. (b) $G = Nk_BT \ln(P\lambda_{\text{th}}^3/k_BT) - k_BT \ln(k_BT/P\Delta V)$. The second term is negligible if $|\ln(\langle V \rangle/N\Delta V)| \ll N$, which holds for any physically reasonable ΔV. (c) $\langle V \rangle = (N+1)k_BT/P \approx Nk_BT/P$. (d) $\text{Var}(V) = k_BT\langle V \rangle \kappa_T$, so $\text{Var}(V)/\langle V \rangle^2 = 1/(N+1)$.
Related: Sections 5.3 and 5.4 of the note. *Source:* [S] 6.12.

Exercise 15: Mixing gases at different temperatures. The monatomic classical ideal gas has the entropy

$$S(U, V, N) = Nk_B \left[\ln\left(\frac{V}{N} \left(\frac{4\pi mU}{3Nh^2} \right)^{3/2} \right) + \frac{5}{2} \right].$$

Two such gases, with equal particle masses, occupy the volumes V_1 and V_2, with the particle numbers N_1 and N_2 and the temperatures T_1 and T_2. The wall between them is removed, and the combined system comes to equilibrium in $V = V_1 + V_2$.

(a) Find the final temperature.

(b) Find ΔS when the two gases are identical and when they are different species.

(c) Show that the difference between the two results is

$$\Delta S_{\text{diff}} - \Delta S_{\text{ident}} = -N k_B \left[x_1 \ln x_1 + x_2 \ln x_2 \right], \qquad x_i = N_i/N,$$

independent of T_1, T_2, V_1 and V_2.

(d) Relate the result of part (c) to $\Delta S_{\text{mix}} \geq 0$ in Section 5.6 of the note and to the ideal-solution model of Section 11.7.

Answer. (a) $T_f = (N_1 T_1 + N_2 T_2)/N$. (b)

$$\Delta S_{\text{ident}} = \tfrac{3}{2} k_B \left[N \ln T_f - N_1 \ln T_1 - N_2 \ln T_2 \right] + k_B \left[N \ln \frac{V}{N} - N_1 \ln \frac{V_1}{N_1} - N_2 \ln \frac{V_2}{N_2} \right],$$

and ΔS_{diff} has $k_B[N_1 \ln(V/V_1) + N_2 \ln(V/V_2)]$ as the second bracket. (d) The difference is the ideal mixing entropy, which the ideal-solution model assumes for the composition dependence of g. *Related:* Sections 5.6 and 11.7 of the note. *Source:* [C] 1045; [S] 5.6.

6 First-Order Phase Transitions and the Unary System

Exercise 16: Critical point of a model fluid.

(a) Below T_c, a model isotherm $P(v)$ has a region with $\partial P/\partial v > 0$. Use the convexity of $f(v)$ in Section 10.3 of the note to explain why this region is unstable. Show that the region shrinks to one point at T_c, so that the critical point satisfies

$$\left(\frac{\partial P}{\partial v} \right)_T = 0, \qquad \left(\frac{\partial^2 P}{\partial v^2} \right)_T = 0.$$

(b) The Dieterici equation of state per particle is $P(v - b) = k_B T \, e^{-a/(k_B T v)}$. Find T_c, v_c and P_c.

(c) In the reduced variables $\tilde{P} = P/P_c$, $\tilde{T} = T/T_c$ and $\tilde{v} = v/v_c$, show that $\tilde{P}(2\tilde{v} - 1) = \tilde{T} \exp[2(1 - 1/\tilde{T}\tilde{v})]$, a law of corresponding states.

(d) Compare the critical ratio $P_c v_c / k_B T_c$ with the van der Waals value $3/8$ and with the measured values of about 0.29 for simple gases.

Answer. (a) $\partial P/\partial v = -\partial^2 f/\partial v^2$, which must be ≤ 0. At T_c, the minimum and the maximum of $P(v)$ merge into an inflection point. (b) $T_c = a/(4k_B b)$, $v_c = 2b$ and $P_c = a/(4e^2 b^2)$. (d) $P_c v_c / k_B T_c = 2/e^2 = 0.271$.
Related: Section 10.3 of the note. *Source:* [S] 8.2.

Exercise 17: Van der Waals coexistence near the critical point. The reduced van der Waals equation of state is $\tilde{P} = 8\tilde{T}/(3\tilde{v} - 1) - 3/\tilde{v}^2$.

(a) Show that the critical isotherm satisfies $\tilde{P} - 1 \approx -\frac{3}{2}(\tilde{v} - 1)^3$.

(b) On the critical isochore $\tilde{v} = 1$ above T_c, show that $P_c \kappa_T = [6(\tilde{T} - 1)]^{-1}$.

(c) Below T_c, apply Maxwell's equal-area rule from Section 10.3 of the note to show that $\tilde{v}_g - \tilde{v}_l \approx 4(1 - \tilde{T})^{1/2}$.

(d) Use the Clapeyron equation to show that the latent heat per particle vanishes as $\ell \approx 6 k_B T_c (1 - \tilde{T})^{1/2}$.

Answer. (c) A numerical Maxwell construction gives $(\tilde{v}_g - \tilde{v}_l)/(1 - \tilde{T})^{1/2} = 4.12$, 4.01 and 4.001 for $1 - \tilde{T} = 10^{-2}$, 10^{-3} and 10^{-4}. (d) $d\tilde{P}/d\tilde{T} \to 4$ on the coexistence curve at T_c, and $P_c v_c = \frac{3}{8} k_B T_c$. The latent heat and Δv vanish together at T_c, as Section 10.3 of the note states.
Related: Sections 10.2 and 10.3 of the note. *Source:* [D] 5.8.

Exercise 18: Latent heat along the vaporization curve. A liquid is incompressible and does not expand with temperature. The vapor volume per particle v_g is much larger than v_l.

(a) Differentiate $\ell = T(s_g - s_l)$ along the coexistence curve. Use a Maxwell relation and the Clapeyron equation to show that

$$\frac{d\ell}{dT} = \frac{\ell}{T} + \Delta c_P - \frac{\ell}{v_g}\left(\frac{\partial v_g}{\partial T}\right)_P, \qquad \Delta c_P = c_{P,g} - c_{P,l}.$$

(b) For an ideal-gas vapor and a constant Δc_P, show that $\ell = \ell_0 + \Delta c_P \, T$.

(c) A saturated vapor expands adiabatically. Show that some liquid condenses if

$$c_{P,l} + T \, \frac{d}{dT}\left(\frac{\ell}{T}\right) < 0.$$

(d) Check the condition for water at 100 °C, with $c_{P,l} = 75.3$, $c_{P,g} = 33.6$ J mol^{-1} K^{-1} and $\ell = 40.7$ kJ mol^{-1}.

Answer. (a) $(\partial s/\partial P)_T = -(\partial v/\partial T)_P$ and $dP/dT = \ell/(Tv_g)$. (c) The entropy of the saturated vapor along the curve is $s_l + \ell/T$, with $ds_l/dT = c_{P,l}/T$. If this entropy decreases with T, an adiabat that starts on the curve enters the two-phase region on cooling. (d) The left side equals -75 J mol^{-1} K^{-1}, so expanding water vapor condenses, as in cloud formation.
Related: Sections 4.5, 10.2 and 10.3 of the note. *Source:* [S] 8.8.

Exercise 19: Boiling point and altitude. An isothermal atmosphere at $T_A = 300.15$ K consists of an ideal gas with the molar mass 30 g mol^{-1}, and the pressure at sea level is P_0. A liquid with the molar latent heat 1000 cal mol^{-1} boils at 105 °C at sea level and at 95 °C on the top of a mountain.

(a) Use the Boltzmann distribution of a molecule in the gravitational potential mgh to find $P_A(h)$.

(b) Integrate the Clausius–Clapeyron equation to find the vapor pressure of the liquid $P_{\text{vap}}(T)$, with $P_{\text{vap}}(378.15\text{ K}) = P_0$.

(c) Find the height of the mountain.

Answer. (a) $P_A = P_0 \exp(-Mgh/RT_A)$. (b) $P_{\text{vap}} = P_0 \exp[-(\ell_m/R)(1/T - 1/378.15\text{ K})]$, with the molar latent heat ℓ_m. (c) $h = (\ell_m T_A/Mg)(1/368.15 - 1/378.15)$ K^{-1} $= 307$ m.
Related: Sections 3 and 10.3 of the note. *Source:* [C] 5955.

Exercise 20: Cooling by solidification in helium-3. Below about 1 K, the entropy per particle of liquid ^{3}He is $s_l = (\pi^2/2)k_B T/T_F$ with $T_F \approx 5$ K. In the solid, the nuclear spins 1/2 are disordered and independent, so that $s_s = k_B \ln 2$. The solid is denser, and $\Delta v = v_l - v_s > 0$ is independent of T.

(a) Find the temperature T^* at which the two phases have equal entropy.

(b) Use the Clapeyron equation to show that the melting curve is

$$P(T) = P^* + \frac{\pi^2 k_B}{4T_F\,\Delta v}\,(T - T^*)^2,$$

with a minimum at T^*. Explain why the liquid is the low-entropy phase below T^*.

(c) State the order of the transition and find the jump in the heat capacity per particle.

(d) The liquid at $T_0 < T^*$ is compressed adiabatically. Assume that T_F does not depend on the pressure. Show that T stays at T_0 until the melting curve is reached.

(e) Further compression converts liquid to solid at fixed total entropy. Show that the system moves down the melting curve to lower T, with the solid fraction $n_s(T) = (T_0 - T)/(T^* - T)$.

Answer. (a) $T^* = (2\ln 2/\pi^2)\,T_F = 0.70$ K for $T_F = 5$ K (measured: 0.32 K). (b) $s_l - s_s = (\pi^2 k_B/2T_F)(T - T^*)$. (c) First order, with $\Delta c = (\pi^2/2)k_B T/T_F$ and $c_s = 0$. (e) The solid has the larger entropy below T^*, so freezing at fixed S absorbs heat from the liquid. This effect is Pomeranchuk cooling.
Related: Sections 10.2 and 10.3 of the note. *Source:* [C] 4552.

7 Binary and Ternary Phase Diagrams

Exercise 21: Compound with a eutectic side and a peritectic side. The system A–B forms the congruently melting compound A_xB_y, denoted γ. Alloys between A and A_xB_y are of the eutectic type, and alloys between A_xB_y and B are of the peritectic type.

(a) Draw the phase diagram.

(b) In the reaction $L + \beta \to \gamma$, which phase lies in the middle of the isotherm? Use this order to decide whether T_P lies above or below the melting point T_γ of the compound.

(c) Use the Gibbs–Konovalov relation to find the slope of the liquidus at the congruent point.

(d) The eutectic point moves slightly to the left or to the right. Describe the change in the total amounts of α and γ and in the amounts of primary crystals and eutectic for an alloy between X_α and X_γ.

Answer. (b) γ lies in the middle, so the liquid lies between X_γ and the γ boundary, which requires $T_P > T_\gamma$. (c) The liquidus is horizontal at X_γ, a horizontal inflection between E and P. (d) The total fractions $n_\alpha = (X_\gamma - X_0)/(X_\gamma - X_\alpha)$ do not change. For a hypoeutectic alloy, a shift to the right gives more primary α, $(X_E - X_0)/(X_E - X_\alpha)$, and less eutectic. A shift to the left has the opposite effect.
Related: Sections 11.4, 11.6, 11.10 and 11.11 of the note. *Source:* Quiz 1, Problem 1.

Exercise 22: Isothermal diffusion in an isomorphous system. In Fig. 1, a bar of A and a bar of B are joined and heated to T_2.

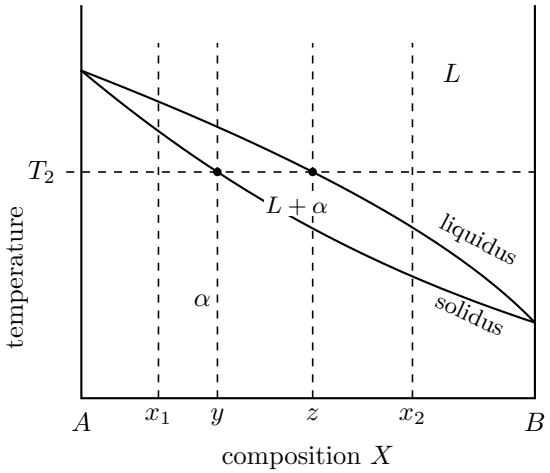


Figure 1: Isomorphous system of Exercise 22.

(a) Which phase is pure B at T_2?

(b) Why can no two-phase layer form in the diffusion couple? Sketch the composition profile during diffusion.

(c) Which phases remain, and with which compositions, for the average compositions x_1 and x_2?

Answer. (a) Liquid, because T_2 lies above the melting point of B. (b) Both chemical potentials are uniform across a two-phase mixture, so no flux crosses the mixture. The profile runs from 0 to y in solid α, jumps from y to z at the interface, and runs from z to 1 in the liquid. (c) For $x_1 < y$, a single solid α of composition x_1. For $x_2 > z$, a single liquid of composition x_2.
Related: Sections 11.4, 11.8 and 11.11 of the note. *Source:* Quiz 1, Problem 2.

Exercise 23: From a P–T projection to a T–X section. Figure 2 shows the P–T projection of a binary system with a eutectoid-type quadruple point Q.

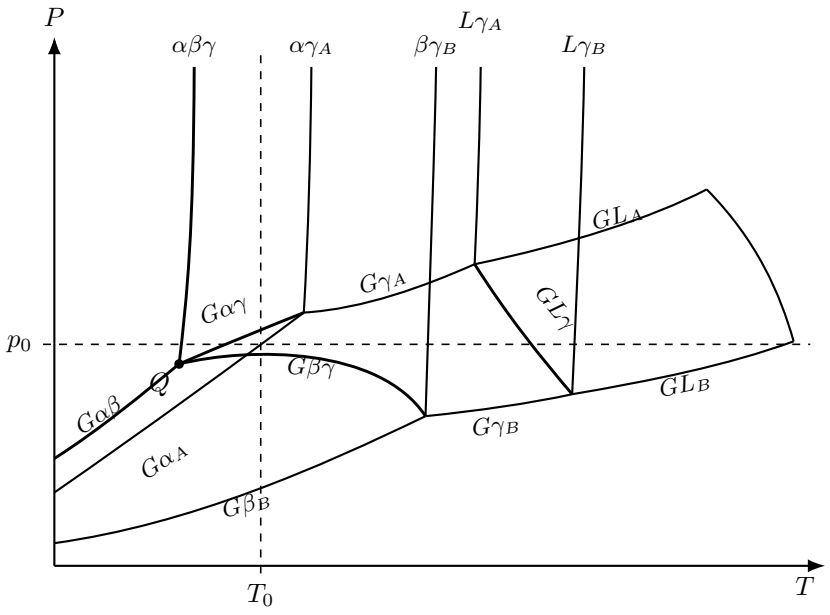


Figure 2: P–T projection of Exercise 23.

(a) List the curves that the isobar p_0 crosses, in order of increasing temperature. Which are binary three-phase equilibria, and which belong to a pure component?

(b) For each three-phase isotherm, find the middle phase and the reaction on cooling.

(c) Draw the T–X diagram at p_0.

Answer. (a) $\alpha\beta\gamma$, $G\alpha\gamma$, $G\alpha_A$, $\beta\gamma_B$, $GL\gamma$, $L\gamma_B$ and GL_B. The curves $\alpha\beta\gamma$, $G\alpha\gamma$ and $GL\gamma$ are binary. Pure A sublimes directly at p_0. (b) $\gamma \to \alpha + \beta$, $G \to \alpha + \gamma$ and $L \to G + \gamma$, all of the eutectic type. (c) The two-phase regions $\alpha + G$, $\gamma + \beta$, $L + \gamma$ and $G + L$ close at the transition points of the pure components. See `quiz1.pdf`, Fig. 7.
Related: Sections 11.5, 11.12 and 11.13 of the note. *Source:* Quiz 1, Problem 3.

Exercise 24: Three-phase reactions in the Ag–Pr system. List all the three-phase reactions with liquid in Fig. 3, with the type, the temperature and the liquid composition of each reaction.

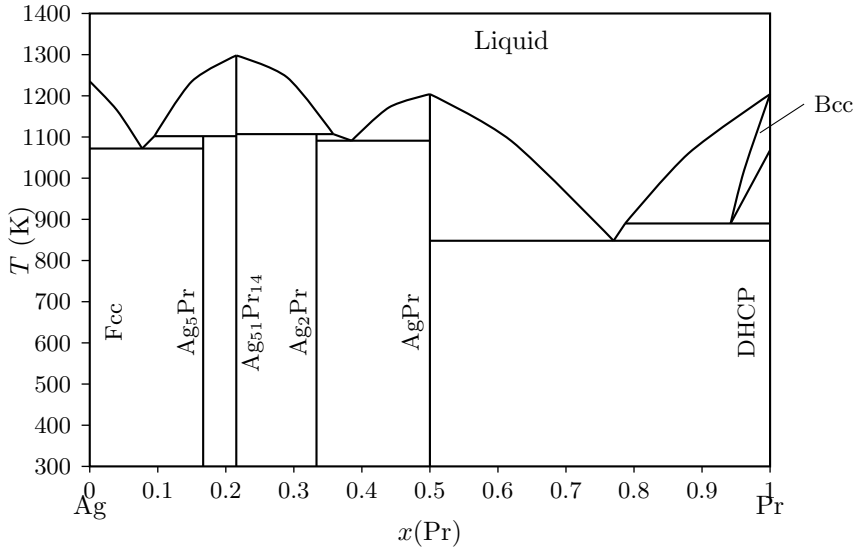


Figure 3: Ag–Pr phase diagram of Exercise 24.

Answer. $L \to$ Fcc + Ag_5Pr (eutectic, 1072 K, $x_L = 0.08$). L+ $Ag_{51}Pr_{14} \to Ag_5Pr$ (peritectic, 1102 K, 0.10). L+ $Ag_{51}Pr_{14} \to Ag_2Pr$ (peritectic, 1107 K, 0.36). $L \to Ag_2Pr$ + AgPr (eutectic, 1091 K, 0.39). Bcc $\to L$+ DHCP (metatectic, 890 K, 0.79). $L \to$ AgPr + DHCP (eutectic, 848 K, 0.77). The congruent melting of $Ag_{51}Pr_{14}$ and AgPr involves only two phases.
Related: Sections 11.5 and 11.10 of the note, Table 5. *Source:* Quiz 1, Problem 4.

Exercise 25: The diopside–anorthite–albite system. Figure 4 shows the Di–An–Ab system, with binary eutectics on the Di–An and Di–Ab sides and a solid-solution loop without a minimum on the Ab–An side.

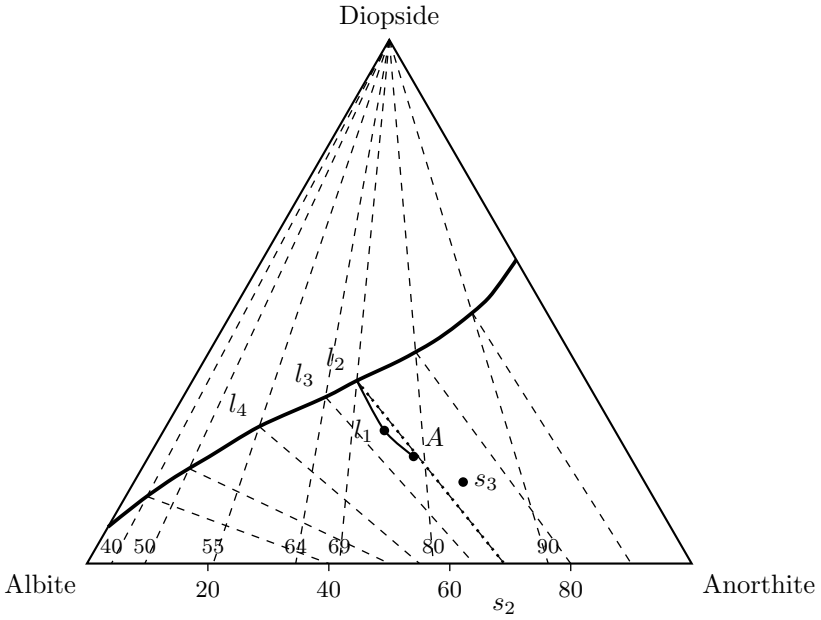


Figure 4: Di–An–Ab system of Exercise 25 (wt%).

(a) Mark the eutectic points. Why does the system have no ternary eutectic?

(b) In which directions does the temperature fall?

(c) State the meaning of the thick curve, the dashed lines from Di, the dashed lines to the Ab–An side, the curve through l_1 and the dotted line.

(d) Which three phases coexist along the cotectic, and which tie triangle belongs to l_4? Draw the isotherm through l_4.

(e) Read the composition of A, and trace the path of the liquid during equilibrium cooling of A. Where does crystallization end, and which plagioclase remains?

Answer. (a) E_1 at 58 wt% Di on the Di–An side and E_2 at 7 wt% Di on the Di–Ab side. The Ab–An loop has no minimum. (b) Along the cotectic from E_1 to E_2, along the Ab–An side from An to Ab, and in each primary field toward the cotectic. (c) Cotectic (L + Di + Pl), diopside–liquid tie lines, liquid–plagioclase tie lines labeled by the An content of the plagioclase, path of the liquid, and the tie line l_2–A–s_2. (d) L + Di + Pl. At l_4, the triangle Di–l_4–An_{55}. The isotherm has two branches that meet at l_4 in a V pointing up the cotectic. (e) A = 20 wt% Di, 44 wt% An, 36 wt%

Ab. The path runs $A \to l_1 \to l_2$ with plagioclase alone, then along the cotectic $l_2 \to l_3 \to l_4$ with diopside and plagioclase. Crystallization ends at l_4 with plagioclase An_{55}.
Related: Sections 11.4 and 11.5 of the note (ternary systems lie beyond the note). *Source:* Quiz 1, Problem 5.